MZ.510 Assignment 2

Due: Wednesday, November 4, 2026 by 23:59

Please email your report (word or pdf) to M. Cagri Gurbuz (mgurbuz@zlc.edu.es)

Problem 1 (20 points):

Speedy Delivery provides two-day delivery service of large parcels across the United States. Each morning at each collection center, the parcels that have arrived overnight are loaded onto several trucks for delivery throughout the area. Since the competitive battlefield in this business is speed of delivery, the parcels are divided among the trucks according to their geographical destinations to minimize the average time needed to make the deliveries.

On this particular morning, the dispatcher for the Blue River Valley Collection Center, Sharon Lofton, is hard at work. Her three drivers will be arriving in less than an hour to make the day's deliveries. There are 9 parcels to be delivered, all at locations many miles apart. As usual, Sharon has loaded these locations into her computer. She is using her company's special software package, a decision support system called Dispatcher. The first thing Dispatcher does is use these locations to generate a considerable number of attractive routes for the individual delivery trucks. These routes are shown in the following table (where the numbers in each column indicate the order of the deliveries), along with the estimated time required to traverse the route.

	Attractive Possible Route									
Delivery Location	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*
A	1				1				1	
B		2		1		2			2	2
C			3	3			3		3	
D	2					1		1		
E			2	2		3				
F		1			2					
G	3						1	2		3
H			1		3					1
I		3		4			2			
Time (in hours)	*6*	*4*	*7*	*5*	*4*	*6*	*5*	*3*	*7*	*6*

Dispatcher is an interactive system that shows these routes to Sharon for her approval or modification. (For example, the computer may not know that flooding has made a particular route infeasible.) After Sharon approves these routes as attractive possibilities with reasonable time estimates, Dispatcher next formulates and solves a integer programming model for selecting three routes that minimize their total time while including each delivery location on exactly one route.

Using the data in the table, (i) demonstrate how Dispatcher can formulate and (ii) solve this model on a spreadsheet (Excel solver).

Problem 2 (15 points):

The Supersuds Corporation is developing its marketing plans for next year's new products. For three of these products, the decision has to be made to purchase a total of five TV spots for commercials on national television networks. Each spot will feature a single product. There, the problem is how to allocate the five spots to these three products, with a maximum of three spots for each product.

The table below shows the estimated impact of allocating zero, one, two, or three spots to each product. This impact is measured in terms of the profit from the additional sales that would result from the spots, considering also the cost of producing the commercial and purchasing the spots. The objective is to allocate five spots to the products so as to maximize the total profit.

	Profit (millions)		
Number of TV spots	*Product 1*	*Product 2*	*Product 3*
0	0	0	0
1	1	0	-1
2	3	2	2
3	3	3	4

(i) *Formulate this problem as an integer program*
(ii) *Solve it using Excel.*

Problem 3 (15 points):

A company is considering opening warehouses in four cities: New York, Los Angeles, Chicago, and Atlanta. Each warehouse can ship 100 units per week. The weekly fixed cost of keeping each warehouse open is $400 for New York, $500 for Los Angeles, $300 for Chicago, and $150 for Atlanta. Region 1 of the country requires 80 units per week, region 2 requires 70 units per week, and region 3 requires 40 units per week. The costs (including production and shipping costs) of sending one unit from a plant to region are shown in the table below.

	To		
From	*Region 1*	*Region 2*	*Region 3*
New York	20	40	50
Los Angeles	48	15	26
Chicago	26	35	18
Atlanta	24	50	35

We want to meet weekly demands at a minimum cost subject to the following restrictions:

1) If the New York warehouse is opened, then the Los Angeles warehouse must be opened.
2) At most two warehouses can be opened.
3) Either the Atlanta or Los Angeles warehouse must be opened.

Formulate an IP that can be used to minimize the weekly costs of meeting demand. (no need to solve it.)

Problem 4 (20 points):

The Polly Esther Cosmetics Company has m manufacturing plants, where S_i is the annual productive capacity of Plant i (assume the company makes a single product). Polly Esther must supply D_j units of product to its j th customer (or demand area), and has n such end-demand points. The company is investigating p possible locations for distribution centers, and it requires that each demand point be served from only one distribution center. Also, assume that the company desires to open exactly r distribution centers.

The following costs data is known to the company:

- c_{ikj}: shipping cost per unit from Plant i to Demand Area j through Distribution Center k
- F_k: fixed operating cost for operating Distribution Center k, if selected.
- f_k: variable cost per unit shipped through Distribution Center k.

Formulate an IP to minimize the relevant costs.

Problem 5 (30 points):

Please read the attached case (Integrating Network Design Models for a Global Supply Network) and answer the questions in Section 3 of the case.